# Nahid Zeinali, PhD

**CLINICAL NLP · LARGE LANGUAGE MODELS · HEALTHCARE DATA SCIENCE**

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Portfolio · GitHub · LinkedIn · Google Scholar

## RESEARCH SUMMARY

Biomedical informatics researcher working at the intersection of clinical natural language processing, large language models, and trustworthy AI for healthcare. Research program develops domain-adaptive language models that turn unstructured clinical and biomedical text into structured, actionable knowledge—**Symptom-BERT** and **Symptom-GPT** for cancer symptom detection across 1M+ EHR notes, **Spiritual-BERT** for holistic care assessment across 3.6M+ clinical narratives, and the open-source **DMP Chef** and **DMP Bridge** systems that orchestrate multiple LLMs to produce FAIR, machine-actionable biomedical documents—together with reusable evaluation frameworks for benchmarking, bias detection, and human-in-the-loop review.

NLM T15 postdoctoral fellow at the University of Utah; author of 16 peer-reviewed publications in informatics, clinical, and AI venues; Teaching experience in informatics and Python programming, with a record of mentoring graduate students through co-authored publications and supervising high-school research interns. Prior career as a healthcare software engineer building HL7-interoperable clinical information systems deployed hospital-wide and training 1,500+ clinicians.

## EDUCATION

**Postdoctoral Fellow (NLM T15 Trainee)** — University of Utah, Salt Lake City, UT — Present

Supervisor: Dr. Guilherme Del Fiol

**Ph.D., Informatics** — University of Iowa, Iowa City, IA — 2025

Dissertation: *Enhancing Cancer Symptom Detection in Electronic Health Records Using Artificial Intelligence*

Advisors: Dr. Stephanie Gilbertson-White (Dissertation Chair); Dr. Weiguo Fan, Professor and Chair in Business Analytics

**M.Sc., Health Informatics** — University of Iowa, Iowa City, IA — 2023

**M.Sc., Medical Informatics** — Tarbiat Modares University, Tehran, Iran — 2016

**B.S., Computer Software Engineering** — Najafabad Azad University, Isfahan, Iran — 2010

## RESEARCH & PROFESSIONAL EXPERIENCE

**Postdoctoral Fellow (NLM T15 Trainee)** — Nov 2026 – Present

*Department of Biomedical Informatics, University of Utah · Salt Lake City, UT*

Supervisor: Dr. Guilherme Del Fiol · GARDE-Chat team

- Adapting state-of-the-art large language models to extract untapped cancer family history from unstructured clinical notes in the electronic health record (EHR).

**AI Research Scientist** — Feb 2025 – Oct 2026

*California Medical Innovation Institute (CalMI2), FAIR Data Innovations Hub · San Diego, CA*

- Designed and built **DMP Chef**, a RAG pipeline with multi-LLM routing (GPT-4, Gemini, Llama) that generates structured biomedical documents from unstructured input, incorporating automated evaluation (SBERT cosine similarity, ROUGE) and human review surveys for continuous quality improvement.
- Created **DMP Bridge**, an open-source tool that converts semi-structured PDF documents into machine-actionable, FAIR-compliant structured JSON, including schema design, document parsing, field-level extraction, and content generation for incomplete fields.

- Developed a reusable **LLM evaluation framework** with gold-standard benchmarking, statistical testing, task-based performance analysis, and bias detection, establishing reproducible quality standards across multiple models.
- Co-developed **Posters-Science,** a web application that transforms scientific posters into machine-readable, structured research objects with a reusable metadata schema adopted by the broader research team.
- Co-developed FAIR-aligned data-standardization pipelines for ophthalmology imaging datasets (**Eye ACT**), including a machine-readable JSON schema, improving ML readiness, interoperability, and reuse of shared imaging data.
- Co-developed a lightweight multimodal classifier that identifies scientific posters **(Poster Sentry)** in open repositories by combining visual and textual signals, enabling automated curation of poster content at scale.

**Research and teaching assistant (Clinical NLP & LLMs)** Aug 2021 – Jan 2025

*University of Iowa · Iowa City, IA*

*Research Assistant, Department of Computer Science and Informatics, University of Iowa*

- Developed **Symptom-BERT** and **Symptom-GPT**, domain-adaptive transformer models for cancer symptom NER across 1M+ oncology EHR notes, including SQL-based cohort extraction, data-pipeline design, and a modular architecture reused across model variants. Symptom-BERT was selected for "Year in Review" by the AMIA Nursing Informatics Working Group.
- Built **Spiritual-BERT**, a domain-adaptive BERT model for spiritual care assessment across 3.6M+ EHR narratives, including a two-iteration annotation schema, inter-annotator agreement validation, and reproducible training pipelines.
- Co-designed **OASIS**, an NLP-driven clinical decision support tool that translates research models into a deployable oncology symptom-monitoring application with a clinician-facing interface for human–AI collaboration in community health settings.
- Conducted a comparative analysis of BERT and GPT architectures for clinical NER (AMIA 2024), establishing best practices for domain-adaptive pre-training and fine-tuning in healthcare.
- Collaborated on a predictive-modeling study (logistic regression, random forest, XGBoost) of 8,156 patient EHR records for 12 cancer-related symptoms, with rigorous pipeline design, leakage prevention, and stratified evaluation (*JAMIA Open*).
- Led a systematic review synthesizing machine-learning approaches to cancer symptom prediction, from classical models through LLMs (*JMIR Cancer*).

*Teaching Assistant, Department of Computer Science and Informatics, University of Iowa*

- Supported instruction in *Introduction to Informatics* and *Python Programming* by grading assignments, providing written feedback, holding office hours, leading lab and discussion sessions, and maintaining course websites.
- Mentored graduate students in natural language processing (NLP), deep learning, and large language model (LLM) development, guiding projects from problem formulation through implementation and co-authored publications.

**NLP Data Scientist** Jun 2024 – Aug 2024

*National Cancer Institute / Frederick National Laboratory, NIH · Frederick, MD*

- Collaborated on NLP- and LLM-based tools for the **RARe-SOURCE** platform, enabling biomedical literature intelligence and semantic search across rare-disease resources for clinical and research use.

**Software Engineer, Healthcare IT** Dec 2016 – Jul 2021

*Khorshid Hospital & Parisian Institute · Iran*

- Designed and built clinical information systems (EMR, Pharmacy IS, Laboratory IS) with HL7-based interoperability, database architecture, and reporting dashboards for hospital-wide clinical and administrative operations.
- Led requirements gathering with clinicians, nurses, and administrators to translate clinical workflows into software specifications and user interfaces.
- Deployed production systems across hospital departments, with responsibility for reliability, data integrity, and version-controlled updates.
- Trained 1,500+ healthcare providers on clinical IT systems, conducting structured usability assessments and incorporating feedback into subsequent releases.

**Computer Instructor** Aug 2010-Jul 2012

*Institute for the Intellectual Development of Children and Young Adults, Iran*

- Taught computer literacy and programming courses to high school students, introducing fundamental computing concepts and programming skills through hands-on instruction.

## GRANTS, FELLOWSHIPS & FUNDING

- **National Library of Medicine (NLM) T15 Postdoctoral Training Fellowship**, Department of Biomedical Informatics, University of Utah, 2026 – present
- **Collaborative Grant Proposal**, *DMP Bridge & DMP Chef: Open-Source Generative AI for FAIR, Machine-Actionable Data Management Plans* — collaborative project with the California Digital Library and CalMI2; NSF proposal in development (2025 – 2026).
- **Ballard and Seashore Dissertation Fellowship** (competitive, funded), University of Iowa, 2024
- **Student Impact Grant**, University of Iowa, 2024
- **Research and Travel Awards** (GPSG/GSS), University of Iowa, 2021 – 2024

## HONORS & AWARDS

- **Graduate Research Excellence Award** — Mathematics, Engineering, or Physical Sciences, University of Iowa, 2025
- **Selection** for Symptom-BERT, AMIA Nursing Informatics Working Group, 2024

## PUBLICATIONS

### Peer-Reviewed Journal Articles

1. C Stephenson, Y Ji, **N Zeinali**, Kc Britt, A Albashayreh. Detecting Language-Related Disparities in Spiritual Care Using Artificial Intelligence. *Journal of Transcultural Nursing*. 2026;0(0). doi:10.1177/10436596261491239
2. **Zeinali N**, Albashayreh A, Fan W, Gilbertson-White S. Using Large Language Models to Detect Anxiety and Nausea/Vomiting Documentation in Clinical Notes of Patients with Cancer. *CIN: Computers, Informatics, Nursing*. 2025. doi:10.1097/CIN.0000000000001418
3. Albashayreh A, **Zeinali N**, Gusen NJ, Ji Y, Gilbertson-White S. An Informatics Approach to Characterizing Rarely Documented Clinical Information in Electronic Health Records: Spiritual Care as an Exemplar. *Applied Clinical Informatics*. 2025;16(4):1146–1156. doi:10.1055/a-2599-6300
4. Albashayreh A, **Zeinali N**, Hagiwara Y, Gilbertson-White S. Disparities in Advance Directive Completion and Life-Sustaining Treatment Preferences in Older Adults. *Journal of Pain and Symptom Management*. 2025;69(5):e733. doi:10.1016/j.jpainsymman.2025.02.451
5. Gilbertson-White S, Albashayreh A, Ji Y, Bandyopadhyay A, **Zeinali N**, et al. Predictors of Concordance between Patient-Reported and Provider-Documented Symptoms in the Context of Cancer and Multimorbidity. *Applied Clinical Informatics*. 2024;15(5):1130–1139. doi:10.1055/s-0044-1791820
6. Albashayreh A, **Zeinali N**, Gilbertson-White S. Innovating the Detection of Care Priorities in Heart Failure Using Large Language Models. *Innovation in Aging*. 2024;8(Suppl 1):1339. doi:10.1093/geroni/igae098.4272
7. Albashayreh A, Bandyopadhyay A, **Zeinali N**, Zhang MIN, Fan W, Gilbertson-White S. Natural Language Processing Accurately Differentiates Cancer Symptom Information in Electronic Health Record Narratives. *JCO Clinical Cancer Informatics*. 2024;8: e2300235. doi:10.1200/CCI.23.00235
8. **Zeinali N**, Albashayreh A, Fan W, Gilbertson-White S. Symptom-BERT: Enhancing Cancer Symptom Detection in EHR Clinical Notes. *Journal of Pain and Symptom Management*. 2024;68(2):190–198.e1. doi:10.1016/j.jpainsymman.2024.05.015
9. Bandyopadhyay A, Albashayreh A, **Zeinali N**, Fan W, Gilbertson-White S. Using Real-World Electronic Health Record Data to Predict the Development of 12 Cancer-Related Symptoms in the Context of Multimorbidity. *JAMIA Open*. 2024;7(3): ooae082. doi:10.1093/jamiaopen/ooae082

10. **Zeinali N**, Youn N, Albashayreh A, Fan W, Gilbertson-White S. Machine Learning Approaches to Predict Symptoms in People with Cancer: Systematic Review. *JMIR Cancer*. 2024;10(1):e52322. doi:10.2196/52322
11. Nazari E, Aghemiri M, **Zeinali N**, Delaram Z, Mehrabian A, Tabesh H. Application of Big Data Analysis in Healthcare Based on "The 6 Building Blocks of Health Systems" Framework: A Survey. *Dokkyo Journal of Medical Sciences*. 2021;48(1):93–101.
12. **Zeinali N**, Asosheh A, Sougand S. Provide Interoperability Model to Interact in Hospital Information Systems. *Journal of Health and Biomedical Informatics*. 2017;4(1):48–58.
13. Shahmoradi M, Safaei AA, Tajrishi H, Nazari E, Delaram Z, Zarei Z, …, **Zeinali N**, et al. The Common Applications of Social Networks in Healthcare. *Health Information Management*. 2016;13(3):243–248.
14. Shahmoradi M, Tajrishi H, Nazari E, Delaram Z, Zarei Z, **Zeinali N**, et al. The Role of Social Networks in Healthcare: Applications and Limitations. *Journal of Health and Biomedical Informatics*. 2015;2(2):124–128.

### Peer-Reviewed Conference Proceedings

1. **Zeinali N**, Albashayreh A, Fan W, Gilbertson-White S. Comparison of BERT Implementations for Enhanced Cancer Symptoms Extraction from Electronic Health Records. *Proceedings of the IEEE International Conference on Artificial Intelligence for Medicine, Health and Care (AIMHC)*, Laguna Hills, CA, 2024. doi:10.1109/AIMHC59811.2024.00011
2. **Zeinali N**, Asosheh A, Setareh S. The Conceptual Model to Solve the Problem of Interoperability in Health Information Systems. *2016 8th International Symposium on Telecommunications (IST)*, 2016:684–689. doi:10.1109/ISTEL.2016.7881909

## CONFERENCE PRESENTATIONS

1. **Zeinali N**†, Patel B, et al. Evaluating the Performance of Large Language Models in Drafting Data Management Plans. *Bioinformatics Open-Source Conference (BOSC)*, 2026.
2. Patel B, **Zeinali N**, et al. Toward Envision Portal: Designing a FAIR and AI-Ready Framework for Ophthalmic Imaging Sharing and Discovery. *ARVO Annual Meeting*, 2026.
3. **Zeinali N**†, Patel B, et al. Evaluating the Effectiveness of an Open-Source Large Language Model in Drafting NIH Data Management Plans. *SciDataCon 2025 / International Data Week*, 2025.
4. **Zeinali N**†, Gilbertson-White S, et al. Leveraging Large Language Models for Named Entity Recognition of Anxiety and Nausea/Vomiting in Patients with Cancer. *AMIA Informatics Summit*, 2025.
5. **Zeinali N**†, Albashayreh A, et al. Comparison of BERT Implementations for Enhanced Cancer Symptom Extraction from Electronic Health Records. *IEEE AIMHC 2024*, Laguna Hills, CA.
6. **Zeinali N**†, Gilbertson-White S, et al. Advanced Detection of Nausea/Vomiting and Anxiety in Patients with Cancer. *AMIA Annual Symposium*, 2024.
7. Albashayreh A, **Zeinali N**, et al. Leveraging Spiritual-BERT for Characterizing Spiritual Care Documentation in EHRs of Older Adults with Heart Failure. *AMIA Annual Symposium*, 2024.
8. Albashayreh A, **Zeinali N**, et al. Innovating the Detection of Care Priorities in Heart Failure Using Large Language Models. Poster, *GSA Annual Scientific Meeting*, 2024.
9. Albashayreh A, **Zeinali N**, et al. Disparities in Advance Directive Completion and Life-Sustaining Treatment Preferences in Older Adults. Poster, *Annual Assembly of Hospice and Palliative Care*, 2024.

*† Presenting author*

## SELECTED RESEARCH PROJECTS & OPEN-SOURCE SOFTWARE

- **DMP Chef** (RAG pipeline): Open-source retrieval-augmented generation pipeline with multi-LLM routing, automated evaluation (SBERT, ROUGE), and human review for structured biomedical document generation. [Code]
- **DMP Bridge** (document-processing software): Open-source tool for converting semi-structured PDFs to machine-actionable JSON with schema design, field-level extraction, and FAIR data standardization. [Link]
- **Symptom-GPT & Symptom-BERT** (clinical NLP): Domain-adaptive transformer and LLM models for cancer symptom NER across 1M+ oncology EHR notes (F1 0.989). [Code]

- **Spiritual-BERT** (EHR phenotyping) Domain-adaptive clinical NLP for holistic care assessment over 3.6M+ EHR narratives with iterative annotation-schema design. [Code]
- **OASIS** (clinical decision support): NLP-driven oncology symptom-monitoring application with a clinician-facing interface for human–AI collaboration.
- **RARe-SOURCE** (NIH): AI-powered biomedical literature intelligence using LLMs and semantic search for rare diseases. [Link]
- **LLM Evaluation Framework** (trustworthy AI): Reusable methodology for benchmarking AI outputs with automated accuracy metrics (SBERT cosine similarity, ROUGE), bias detection, and human review surveys. [Link]
- **Posters.science** (web application): Transforms scientific posters into machine-readable research objects with a reusable metadata schema. [Link]
- **EyeACT** (FAIR ophthalmology data standardization): FAIR-aligned data-standardization pipelines and a machine-readable JSON schema for ophthalmology imaging datasets, improving ML readiness, interoperability, and reuse. [Link]
- **PosterSentry** (multimodal classification): Lightweight multimodal classifier that identifies scientific posters in open repositories by combining visual and textual signals, enabling automated curation of poster content at scale. [Code]

## TEACHING & MENTORING

### Teaching

- **Teaching Assistant**, University of Iowa — *Introduction to Informatics* and *Python Programming*: graded assignments and provided written feedback, held office hours, led lab/discussion sessions, and maintained the course website.
- Designed and delivered clinical IT training for 1,500+ healthcare providers (Khorshid Hospital & Parisian Institute, 2016 – 2021).
- **Instructor**, computer and programming courses for high-school students, Institute for the Intellectual Development of Children and Young Adults, Iran, 2010 – 2012.

### Mentoring

- Mentored graduate students and junior researchers at the University of Iowa on NLP, deep learning, LLM development, and responsible AI, guiding them from problem framing through co-authored publications.
- Mentored an Iowa high school student in an independent research project on machine learning applications in cancer symptom management, including advanced research methods (2026–present).

## TECHNICAL & PROFESSIONAL EXPERTISE

### Technical

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|---|---|
| **Programming** | Python (expert), R, SQL, JavaScript, Bash, HTML/CSS |
| **LLMs & RAG** | GPT-4, Gemini, Llama, BERT, Clinical BERT, Bio BERT, domain-adaptive pre-training, fine-tuning, retrieval-augmented generation, multi-LLM routing, prompt engineering, SBERT evaluation |
| **AI/ML Frameworks** | PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, XGBoost, spaCy, LangChain |
| **NLP Methods** | Named entity recognition, text classification, domain-adaptive pre-training, clinical text mining, EHR phenotyping, unstructured report analysis |
| **Healthcare Data & Standards** | EHRs, clinical narratives, HL7/FHIR, FAIR data standards, ICD-10, SNOMED CT, DICOM |
| **Software Engineering** | Git/GitHub, Docker, CI/CD, REST APIs, automated testing, version-controlled schema design, modular pipeline architecture |
| **Data & Infrastructure** | SQL, vector databases, database design, data-pipeline architecture, reproducible research, cloud computing |
| **Evaluation & Quality** | LLM benchmarking, SBERT cosine similarity, ROUGE, bias detection, fairness metrics, human-in-the-loop review, usability assessment, statistical testing |

### Academic & Professional

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|---|---|
| **Research Leadership & Funding** | Research program design, grant and proposal development (NIH/NSF), NIH data management and sharing plans, systematic reviews and study design, interdisciplinary collaboration with clinicians and data stewards, peer review (AMIA) |
| **Teaching & Curriculum** | Course instruction and teaching assistance (Introduction to Informatics, Python Programming), curriculum planning and rubric-based assessment, lab and discussion facilitation, course-site administration, professional training design for 1,500+ clinicians |
| **Mentoring & Supervision** | Graduate-student research mentoring from problem framing through co-authored publication, supervision of high-school research interns, peer mentoring |
| **Scientific Communication** | Peer-reviewed manuscript writing (16 publications), conference talks and posters (AMIA, BOSC, IEEE, ARVO, SciDataCon), technical documentation, open-source software release and maintenance |
| **Open Science & Responsible AI** | FAIR data stewardship, machine-actionable metadata and schemas, reproducible open-source workflows, responsible AI governance, fairness and bias assessment, human–AI collaboration in clinical settings |

## PROFESSIONAL SERVICE & AFFILIATIONS

- Reviewer, AMIA Annual Symposium
- Peer Reviewer, AMIA Informatics Summit 2025, American Medical Informatics Association
- Member, BOSC (Bioinformatics Open-Source Conference) and ISMB (Intelligent Systems for Molecular Biology) communities
- Peer-to-Peer Mentor & Mentee Volunteer, University of Iowa, 2024
- Student Volunteer, ISO Organization, University of Iowa, 2021 – 2023